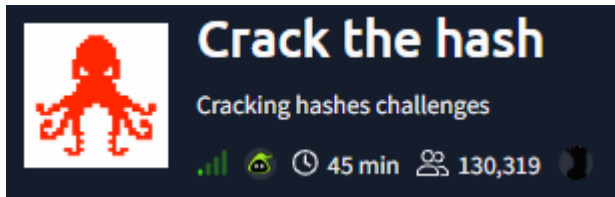


12-18-2025 - THM CTF - Crack The Hash



<https://tryhackme.com/room/crackthehash>

Level: EASY

Tools:

- <https://www.tunnelsup.com/hash-analyzer/>
- <https://crackstation.net/>
- Hashcat

References:

- <https://medium.com/@juwonbal/cracking-the-hash-my-deep-dive-into-tryhackmes-hash-cracking-challenge-ea566f1c1540>
- <https://medium.com/@sunya22839/crack-the-hash-walkthrough-75dea6faeb4a>
- https://hashcat.net/wiki/doku.php?id=example_hashes

Description of the Challenge

This CTF is about decrypting hashes. Some hashing conventions aren't secure and hence are obsolete. The challenge consists of identifying the type of hash a string is and decrypting the hashes using online tools. Task 1 has 5 hashes to be cracked and Task 2 has 4 hashes requiring the use of **HashCat** for cracking. The easiest challenges have been grouped together under **Part 1**. The challenges requiring hashcat are grouped together in **Part 2**

Part 1

Task 1 ○ Level 1

Can you complete the level 1 tasks by cracking the hashes?

Answer the questions below

48bb6e862e54f2a795ffc4e541caed4d

Answer format: ****

Check

?

CBFDAC6008F9CAB4083784CBD1874F76618D2A97

Answer format: *****

Check

?

1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032

Answer format: *****

Check

?

\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom

Answer format: ****

Check

?

279412f945939ba78ce0758d3fd83daa

Answer format: *****

Check

?

I input all of the hashes from Task 1 into CrackStation and this is what I got:

Enter up to 20 non-salted hashes, one per line:

48bb6e862e54f2a795ffc4e541caed4d
CBFDAC6008F9CAB4083784CBD1874F76618D2A97
1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032
\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom
279412f945939ba78ce0758d3fd83daa

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Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
48bb6e862e54f2a795ffc4e541caed4d	md5	easy
CBFDAC6008F9CAB4083784CBD1874F76618D2A97	sha1	password123
1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032	sha256	letmein
\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom	Unknown	Unrecognized hash format.
279412f945939ba78ce0758d3fd83daa	md4	Eternity22

Color Codes: Green: Exact match, Yellow: Partial match, Red: Not found.

Crackstation didn't recognize the fourth hash. This required some deep diving so I used TunnelsUP and got this:

Hash:	\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom
Hash type:	bcrypt
Bit length:	184
Character length:	60
Character type:	\$2x\$x\$ followed by base64
Hash:	eUznr71EeNkJkUlypTsgbX1H68wsRom
Salt:	Dwt1BZj6pcyc3Dy1FWZ5ie

The fourth hash required the use of hashcat, which will be shown in Part 2 of this writeup. So I moved onto the next two challenges from Task 2 and they were very straightforward like in the previous examples:

Hash	Type	Result
F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85	sha256	paule
1DFECA0C002AE40B8619ECF94819CC1B	NTLM	n63umy81kf4i

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

Crackstation made life easy for solving questions 1, 2, 3, 5, 6 & 7.

Task 2

Cracking each hash in Task 2 required these basic steps:

1. Identify the type of hash using CrackStation or TunnelsUP Hash Analyzer
2. Put the hash into its own text file using nano.
3. Use the [Example Hashes list at hashcat's website](#) if the prior two tools fail.
4. Download the rockyou.txt wordlist (the link provided in the Crack the Hash room or the file is in the ls /usr/share/wordlists/ directory in the TryHackMe AttackBox.

5. Using the hints provided in the TryHackMe task sections, write some awk commands that will filter (reducing the keyspace on) the rockyou.txt wordlist for a shorter list based on the amount of characters the answer is supposed to be in the THM answer field.

For example:

```
awk 'length($0)==4' /usr/share/wordlists/rockyou.txt >
rockyou_4.txt
```

Awk 'length(\$0)==4' extracts 4 character words from the rockyou.txt wordlist and compiles all of those words into the rockyou_4.txt wordlist, which will be MUCH smaller (hence *faster*) for hashcat to work from.

6. The next step was cracking the hash using hashcat.

```
hashcat -m 3200 -a 0 hash.txt rockyou_4.txt --force
```

- -m 3200 → bcrypt
 - -m xxxx → is the mode/the hashing method (MD5, SHA1, SHA256)
- -a 0 → dictionary attack
 -

7. Now recover the password:

```
hashcat -m 3200 hash.txt --show
```

For the fourth hash from Task 1, hashcat delivered the answer: bleh.

Now rinse and repeat:

Task 2 Challenge/Question 3

Hash:

\$6\$aReallyHardSalt\$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJML9be.cfi3/qxlf.hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.

Salt: aReallyHardSalt

1. I used the Example Hashes list from hashcat to determine this is a SHA-512 has. So the mode for it will be -m 1800.
2. I put the hash with its salt into the h3.txt file.
3. Wrote the following command to reduce the wordlist to a more manageable and speedier size:

```
awk 'length($0)==6' /usr/share/wordlists/rockyou.txt >
rockyou_6.txt
```

4. Ran the command to perform the crack:

```
hashcat -m 1800 -a 0 h3.txt rockyou_6.txt --force
```

5. Finally:

```
hashcat -m 1800 h3.txt --show
```

And the answer is: waka99

Task 2 Challenge 4

Hash: e5d8870e5bdd26602cab8dbe07a942c8669e56d6

Salt: tryhackme.

1. Identified it as a SHA-1
2. Put the hash and salt into the h4.txt file using nano.
3. Ran the command:

```
hashcat -m 120 -a 0 h4.txt /usr/share/wordlists/rockyou.txt
--force
```

4. Ran the command to show the result:

```
hashcat -m 120 h4.txt --show
```

And the answer is: 481616481616